Frequency Distributions

Chapter 2

Biostatistics and Data Analysis



Chapter Goals

- Frequency Distributions
- Types of Frequency Distribution
- Constructing a Frequency Distribution Table
- Charts and graphs
- Constructing Statistical Graphs



Frequency Distributions

- After collecting data, the first task for a researcher is to organize and simplify the data so that it is possible to get a general and quick overview of the results.
- This is the goal of descriptive statistical techniques.
- One method for simplifying and organizing data is to construct a **frequency distribution**.
- Definition of "**FREQUENCY DISTRIBUTION**"

  "The frequency of a value is the number of times it occurs in a dataset".

Types of Frequency Distribution

- There are three major types for “Frequency Distribution”.

1. **Categorical Frequency Distribution**
2. **Ungrouped Frequency Distribution**
3. **Grouped Frequency Distribution**



Types of Frequency Distribution

1. **Categorical Frequency Distribution**

- A categorical frequency distribution is a table to organize data that can be placed in specific categories.
- It is used when the data is nominal.



Different Classes of Blood Types

Class	Frequency
A	5
B	7
O	9
AB	4

Types of Frequency Distribution

1. **Categorical Frequency Distribution**

- **Example**
- Twenty-five army inductees were given a blood test to determine their blood type.
- **Raw Data:** A,B,B,AB,O O,O,B,AB,B B,B,O,A,O

  A,O,O,O,AB AB,A,O,B,A

Class	Frequency
A	5
B	7
O	9
AB	4

Types of Frequency Distribution

2. Ungrouped Frequency Distribution

- It is used for numerical data when the range of data is small.
- Here, there is no class intervals and the exact frequency is written for the individual data.



x	f
12	5
13	7
14	6
15	4
16	3

Types of Frequency Distribution

- **2. Ungrouped Frequency Distribution**
- **Example**
- Take sample of 16 children from a primary school and get the following data about the number of their decayed teeth / dental caries:

  3,5,2,4,0,1,3,5,2,3,2,3,3,2,4,1

- To construct a frequency table
- We need two columns:
- 1.Variable name
- 2.Frequency (f):

No. of decayed teeth	Frequency
0	1
1	2
2	4
3	5
4	2
5	2
Total	n = 16

Types of Frequency Distribution

3. Grouped Frequency Distribution

- It is used when the range of the data is large.



Range	Frequency
1-10	2
11-20	7
21-30	10
31-40	3
41-50	1

Constructing a Grouped Frequency Distribution

STEP 1 Find Range

STEP 2 Fix class width

STEP 3 Fix class interval.

STEP 4 Find mid interval.

STEP 5 Find frequency

STEP 6 Find Relative Frequency

STEP 7 Find Percentage frequency

STEP 8 Find Cumulative frequency

STEP 9 Find Cumulative Relative Frequency

Constructing a Grouped Frequency Distribution

STEP 1 Find Range

- It is the difference between the largest and the smallest observation in the data set.
- Range = Largest – Smallest

42	26	32	34	57
30	58	37	50	30
53	40	30	47	49
50	40	32	31	40
52	28	23	35	25
30	36	32	26	50
55	30	58	64	52
49	33	43	46	32
61	31	30	40	60
74	37	29	43	54

Smallest

Largest

$$\begin{aligned} \text{Range} &= \text{Largest} - \text{Smallest} \\ &= 74 - 23 \\ &= 51 \end{aligned}$$

Note: *One can order the data values from smallest to largest to help find the range.*

Constructing a Grouped Frequency Distribution

STEP 2 Fix class width

- Find the class width by dividing the range by the number of classes.
- Range = Largest – Smallest

  = 74 – 23 = 51

- Suppose number of classes are 6.
- *Class Width* = *Range*/ Number of classes = 51/6 = 8.5 ≈ 9
- Rounding Rule: Always round up.



Constructing a Grouped Frequency Distribution

STEP 3 Fix class interval

- For convenience sake, we will choose the lowest data value i.e 23, for the first lower class limit.
- The subsequent lower class limits are found by adding the width to the previous lower class limits. (23 + 9 = 32)
- The first upper class limit is one less than the next lower class limit.
- The subsequent upper class limits are found by adding the width to the previous upper class limits. (31 + 9 = 40)

Class Interval
23 - 31
32 – 40
41 – 49
50 – 58
59 – 67
68 – 76

Constructing a Grouped Frequency Distribution

STEP 4 Find mid interval (Midpoint)

- The midpoint of each class interval is called the class midpoint or the class mid interval.
- Class midpoint = $\frac{\text{Class Beginning Point + Class Ending Point}}{2}$

$$= \frac{23+31}{2} = 27$$



Constructing a Grouped Frequency Distribution

STEP 4 Find frequency

- The number of particular values for a class interval, have been observed in the given data.

112	100	127	120	134	118	105	110	109	12
110	118	117	16	118	122	114	114	105	109
107	112	14	115	118	117	118	122	106	110
116	108	110	121	113	120	119	111	104	111
120	13	120	117	105	110	118	112	114	114

Interval	Frequency
0 to 19	4
20 to 39	
40 to 59	
60 to 79	
80 to 99	
100 to 119	

Constructing a Grouped Frequency Distribution

STEP 6 Find Relative Frequency **(Relative frequency = frequency/n)**

- The relative frequency is the proportion of the total frequency that is any given class interval in a frequency distribution.

Class Interval	Frequency	Relative Frequency
20-under 30	6	.12
30-under 40	18	.36
40-under 50	11	.22
50-under 60	11	.22
60-under 70	3	.06
70-under 80	1	.02
	50	1.00

$\frac{6}{50} =$

$\frac{18}{50} =$

Constructing a Grouped Frequency Distribution

STEP 7 Find Percentage frequency

- Percentage frequency = Relative frequency × 100%

Class interval	Frequency	Relative Frequency	Percentage Frequency	Cumulative frequency	Cumulative Relative Frequency
100 - 104	2	0.04	4	2	0.04
105 - 109	8	0.16	16	10	0.2
110 - 114	18	0.36	36	28	0.56
115 - 119	13	0.26	26	41	0.82
120 - 124	7	0.14	14	48	0.96
125 - 129	1	0.02	2	49	0.98
130 - 134	1	0.02	2	50	1

Constructing a Grouped Frequency Distribution

STEP 8 Find Cumulative frequency

- The cumulative frequency is a running total of frequencies through the classes of a frequency distribution.

Class Interval	Frequency	Cumulative Frequency
20-under 30	6	6
30-under 40	18	18 + 6 → 24
40-under 50	11	11 + 24 → 35
50-under 60	11	46
60-under 70	3	49
70-under 80	1	50
Total	50	

Cumulative frequency:

- Cumulative frequency of the 1st class interval = frequency.
- Cumulative frequency of a class interval = frequency + cumulative frequency of the previous class interval

Constructing a Grouped Frequency Distribution

STEP 9 Find Cumulative Relative Frequency

- Cumulative Relative frequency = Cumulative frequency/n

Class interval	Frequency	Cumulative frequency	Cumulative Relative Frequency
100 - 104	2	2	0.04
105 - 109	8	10	0.2
110 - 114	18	28	0.56
115 - 119	13	41	0.82
120 - 124	7	48	0.96
125 - 129	1	49	0.98
130 - 134	1	50	1

Example:-1
Constructing a Grouped Frequency Distribution

The following data represent the record high temperatures for each of the 50 states. Construct a grouped frequency distribution for the data using 7 classes.

112	100	127	120	134	118	105	110	109	112
110	118	117	116	118	122	114	114	105	109
107	112	114	115	118	117	118	122	106	110
116	108	110	121	113	120	119	111	104	111
120	113	120	117	105	110	118	112	114	114

Example:-1
Constructing a Grouped Frequency Distribution

STEP 1 Find Range

STEP 2 Fix class width and fix class interval.

STEP 3 Find mid interval.

STEP 4 Find frequency

STEP 5 Find Relative Frequency

STEP 6 Find Percentage frequency

STEP 7 Find Cumulative frequency

STEP 8 Find Cumulative Relative Frequency

Example:-1
Constructing a Grouped Frequency Distribution

Find the class width by dividing the range by the number of classes 7.

$$Range = High - Low$$
$$= 134 - 100 = 34$$

$$Class\ Width = Range/7 = 34/7 = 4.85 \approx 5$$

Rounding Rule: Always round up.

Class Limits
100 - 104
105 - 109
110 - 114
115 - 119
120 - 124
125 - 129
130 - 134

Example:-1
Constructing a Grouped Frequency Distribution

- For convenience sake, we will choose the lowest data value, 100, for the first lower class limit.
- The subsequent lower class limits are found by adding the width to the previous lower class limits.

Class Limits
100 - 104
105 - 109
110 - 114
115 - 119
120 - 124
125 - 129
130 - 134

- The first upper class limit is one less than the next lower class limit.
- The subsequent upper class limits are found by adding the width to the previous upper class limits.

Example:-1
Constructing a Grouped Frequency Distribution

Class interval	Frequency	Mid interval	Relative Frequency	Percentage frequency	Cumulative frequency	Cumulative Relative Frequency
100 - 104						
105 - 109						
110 - 114						
115 - 119						
120 - 124						
125 - 129						
130 - 134						

Example:-1
Constructing a Grouped Frequency Distribution

Class interval	Frequency	Mid interval	Relative Frequency	Percentage frequency	Cumulative frequency	Cumulative Relative Frequency
100 - 104	2	102	0.04	4	2	0.04
105 - 109	8	107	0.16	16	10	0.2
110 - 114	18	112	0.36	36	28	0.56
115 - 119	13	117	0.26	26	41	0.82
120 - 124	7	122	0.14	14	48	0.96
125 - 129	1	127	0.02	2	49	0.98
130 - 134	1	132	0.02	2	50	1

Example:-2
Constructing a Grouped Frequency Distribution

The following table gives the hemoglobin level (g/dl) of a sample of 50 men (class width-10).

17.0	17.7	15.9	15.2	16.2	17.1	15.7	17.3	**13.5** 16.3
14.6	15.8	15.3	16.4	13.7	16.2	16.4	16.1	17.0
15.9	14.0	16.2	16.4	14.9	17.8	16.1	15.5	**18.3**
15.8	16.7	15.9	15.3	13.9	16.8	15.9	16.3	17.4
15.0	17.5	16.1	14.2	16.1	15.7	15.1	17.4	16.5
14.4	16.3	17.3	15.8					

Example:-2
Constructing a Grouped Frequency Distribution

From frequencies construct a frequency distribution table and Find

- The number of people whose haemoglobin levels are between 17.0 and 17.9 =
- From cumulative frequencies: The number of people whose haemoglobin levels are less than or equal to 15.9 =
- The number of people whose haemoglobin levels are less than or equal to 17.9 =
- From percentage frequencies:The percentage of people whose haemoglobin levels are between 17.0 and 17.9 =
- From cumulative percentage frequencies:The percentage of people whose haemoglobin levels are less than or equal to 14.9 =
- The percentage of people whose haemoglobin levels are less than or equal to 16.9 =

Example:-2
Constructing a Grouped Frequency Distribution

Class interval	Frequency	Cumulative Frequency	Relative Frequency	Cumulative Relative Frequency	Percentage frequency	Cumulative Percentage frequency
13.0-13.9	3	3	0.06	.06	6	6
14.0-14.9	5	8	.1	.16	10	16
15.0-15.9	15	23	.3	.46	30	46
16.0-16.9	14	39	.28	.78	32	78
17.0-17.9	10	49	.2	.98	20	98
18.0-18.9	1	50	.02	1	2	100

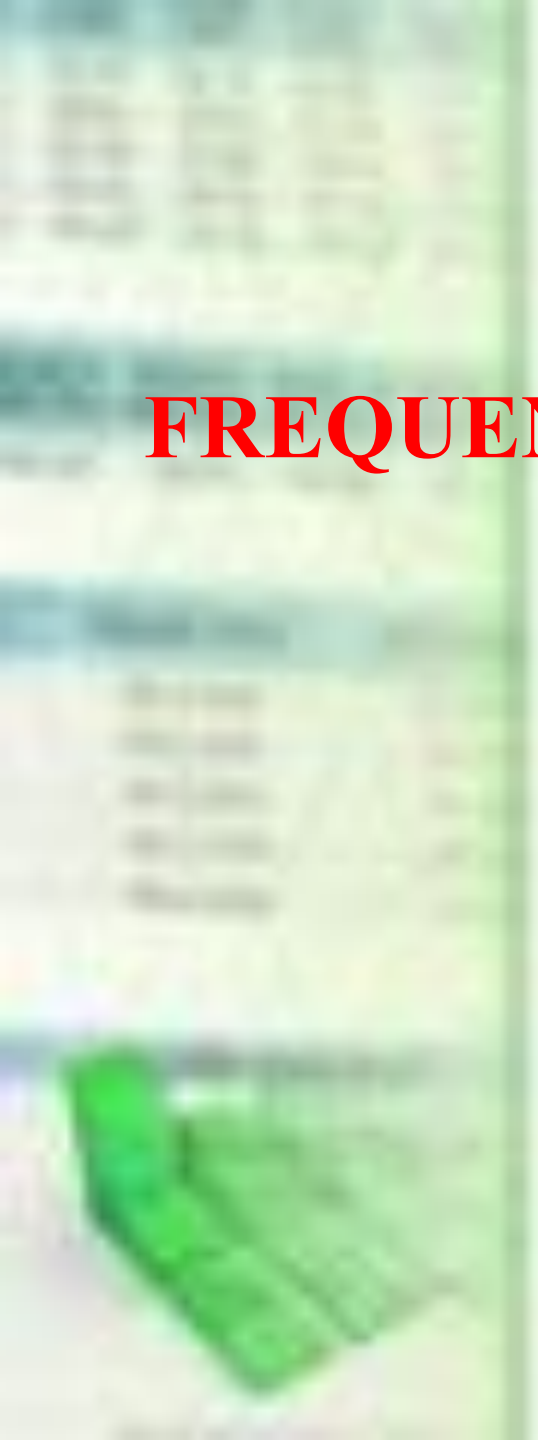
Example:-2
Constructing a Grouped Frequency Distribution

From frequencies:

- The number of people whose haemoglobin levels are between 17.0 and 17.9 = **10**
- From cumulative frequencies:The number of people whose haemoglobin levels are less than or equal to 15.9 = **23**
- The number of people whose haemoglobin levels are less than or equal to 17.9 = **49**
- From percentage frequencies:The percentage of people whose haemoglobin levels are between 17.0 and 17.9 = **20%**
- From cumulative percentage frequencies:The percentage of people whose haemoglobin levels are less than or equal to 14.9 = **16%**
- The percentage of people whose haemoglobin levels are less than or equal to 16.9 = **78%**

Class Interval	Frequency	Cumulative Frequency	Relative Frequency	Cumulative Relative Frequency	Percentage Frequency	Cumulative Percentage Frequency
13.0 – 13.9	3	3	0.06	0.06	6%	6%
14.0 – 14.9	5	8	0.10	0.16	10%	16%
15.0 – 15.9	15	23	0.30	0.46	30%	46%
16.0 – 16.9	16	39	0.32	0.78	32%	78%
17.0 – 17.9	10	49	0.20	0.98	20%	98%
18.0 – 18.9	1	50	0.02	1.00	2%	100%

FREQUENCY DISTRIBUTION GRAPHS



Frequency distribution graphs

- Frequency distribution graphs are useful because they show the entire set of scores.
- At a glance, you can determine the highest score, the lowest score, and where the scores are centered.
- The graph also shows whether the scores are clustered together or scattered over a wide range.



Common Statistical Graphs

- **Histogram** : Vertical bar chart of frequencies
- **Frequency Polygon** : line graph of frequencies
- **Ogive** : line graph of cumulative frequencies
- **Bar graph:** Chart or graph that presents data with rectangular bars

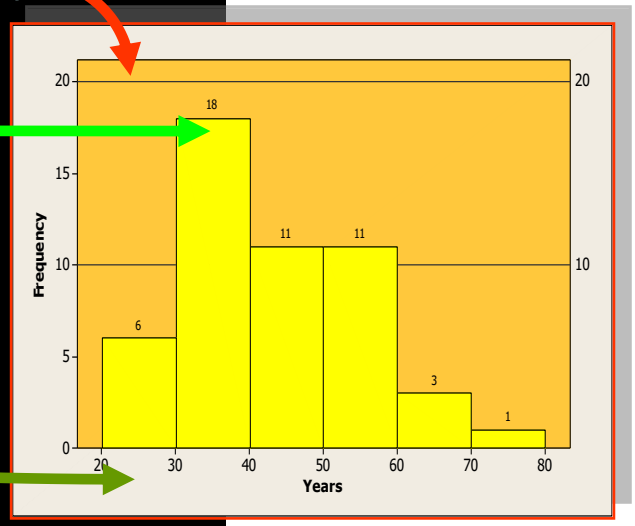
  The bars can be plotted vertically or horizontally.
- **Pie Chart** : Proportional representation for categories of a whole
- **Stem and Leaf Plot:** each data value is split into a "stem" and a "leaf"
- **Scatter Plot:** A scatter plot is a two-dimensional graph plot of pairs of

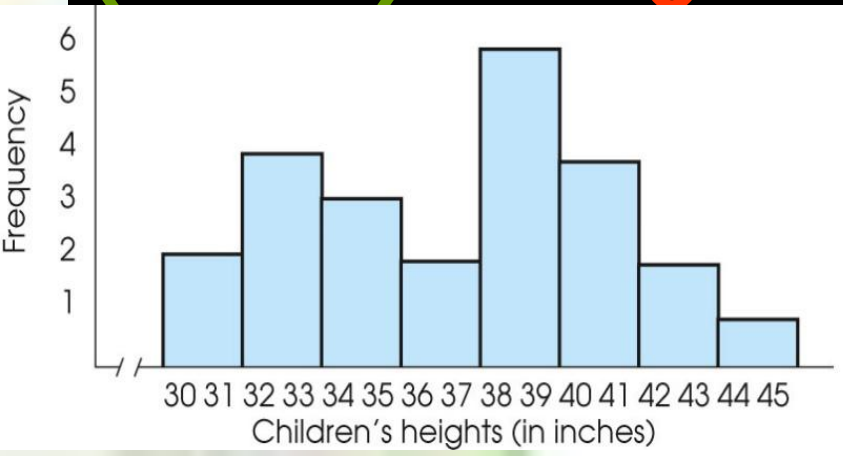
  points from two numerical variables.

Histogram

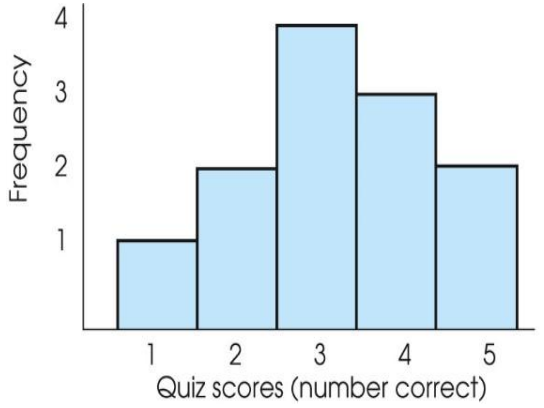
- In a **histogram**, a bar is centered above each score (or class interval) so that the height of the bar corresponds to the frequency.

Class Interval	Frequency
20-under 30	6
30-under 40	18
40-under 50	11
50-under 60	11
60-under 70	3
70-under 80	1





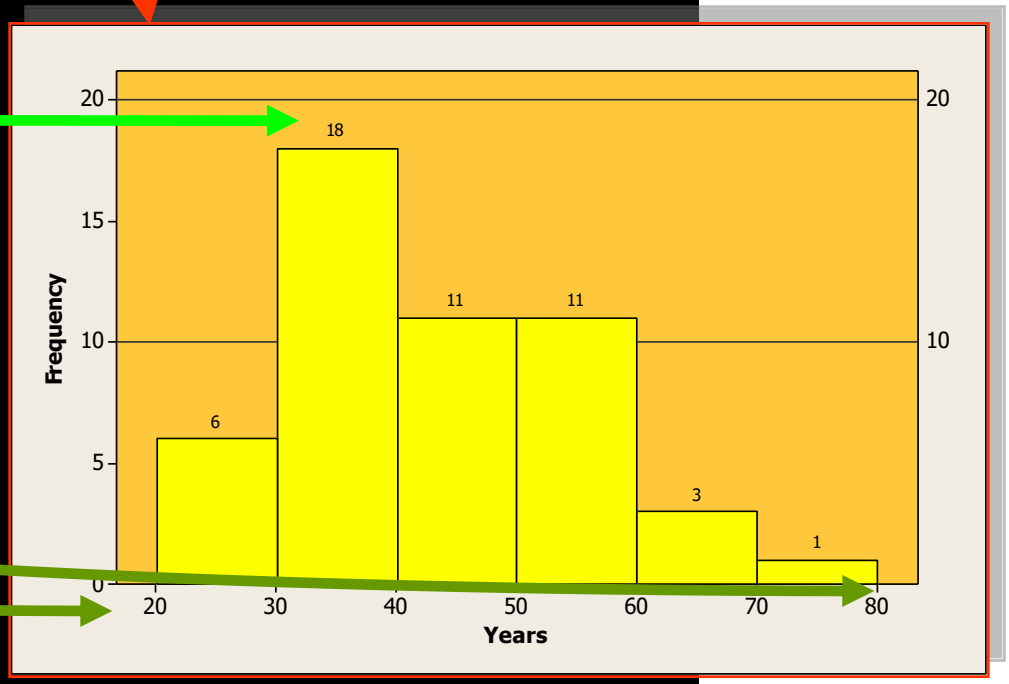
X	f
44–45	1
42–43	2
40–41	4
38–39	6
36–37	2
34–35	3
32–33	4
30–31	2



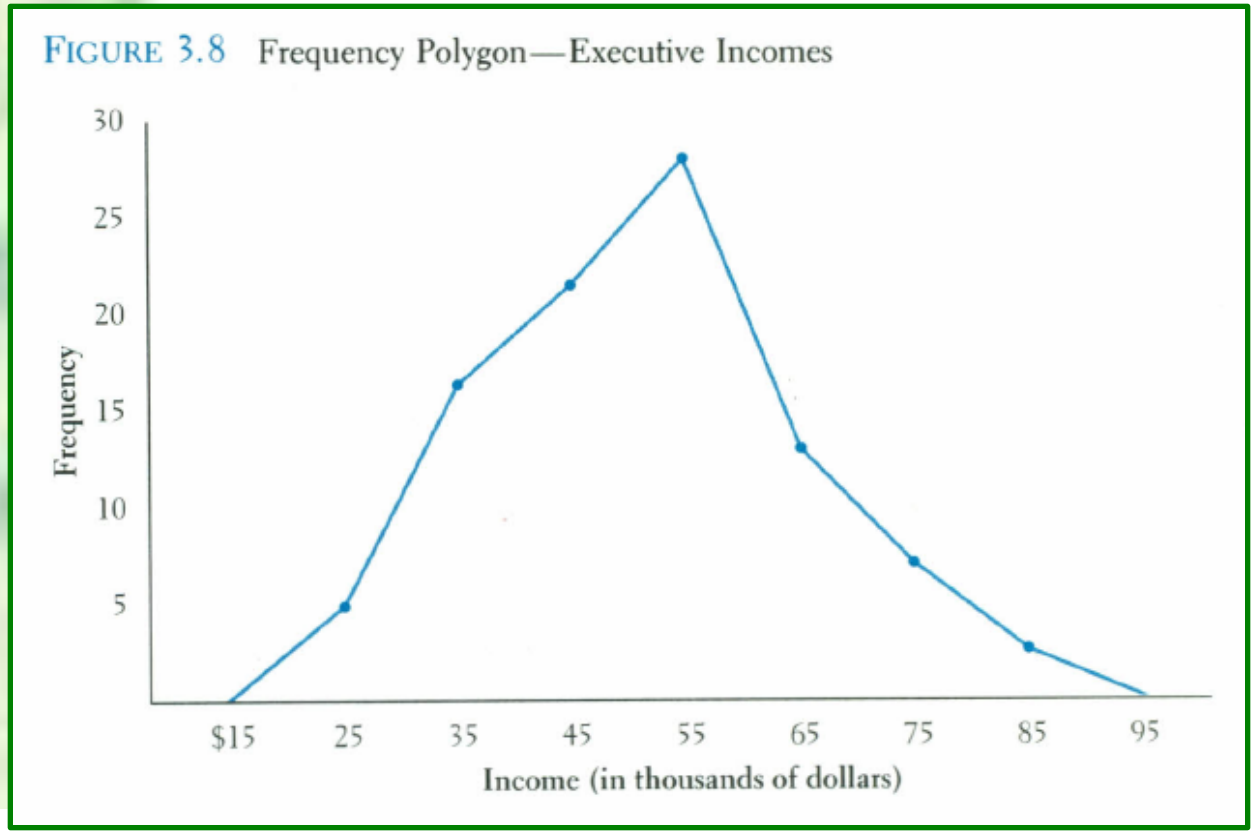
X	f
5	2
4	3
3	4
2	2
1	1

Histogram Construction

Class Interval	Frequency
20-under 30	6
30-under 40	18
40-under 50	11
50-under 60	11
60-under 70	3
70-under 80	1



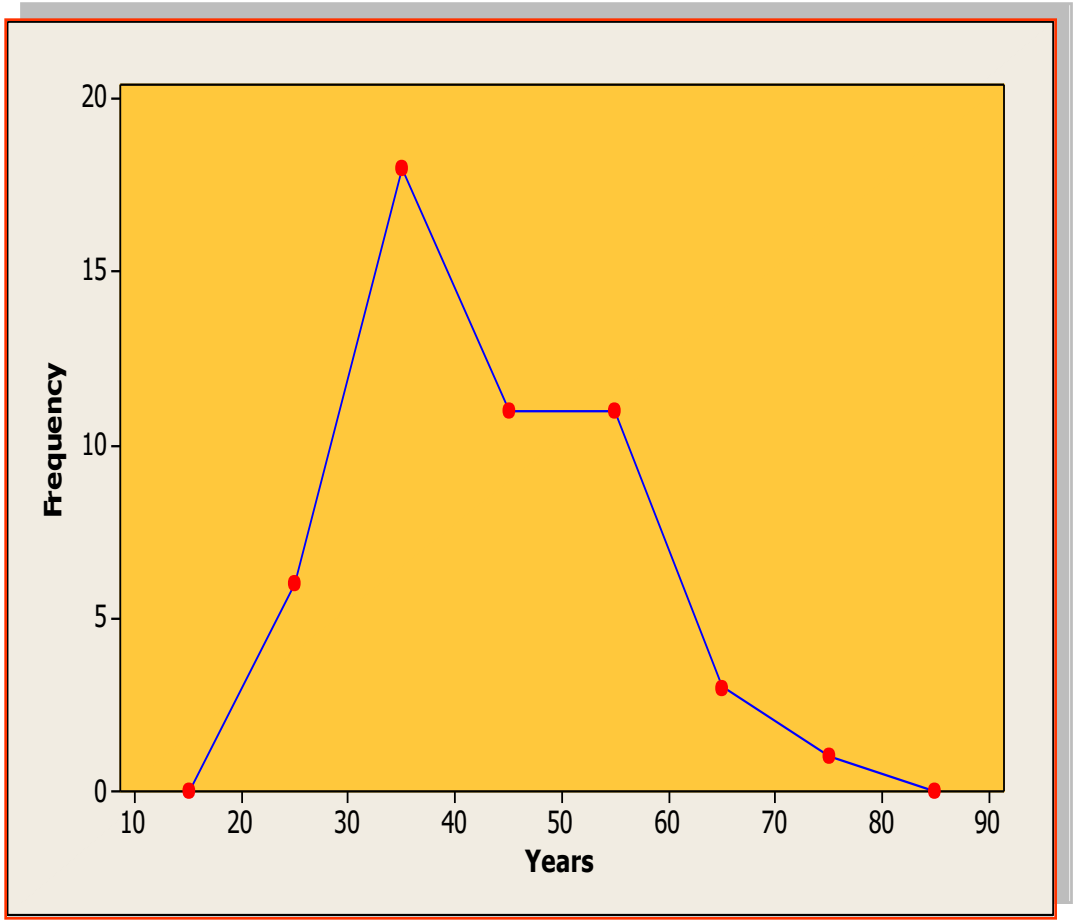
Frequency Polygons



- The **frequency polygon** is a graph that displays the data by **using lines that connect points** plotted for the frequencies at the **class midpoints.**
- The frequencies are represented by the heights of the points.
- The class midpoints are represented on the horizontal axis.

Frequency Polygon

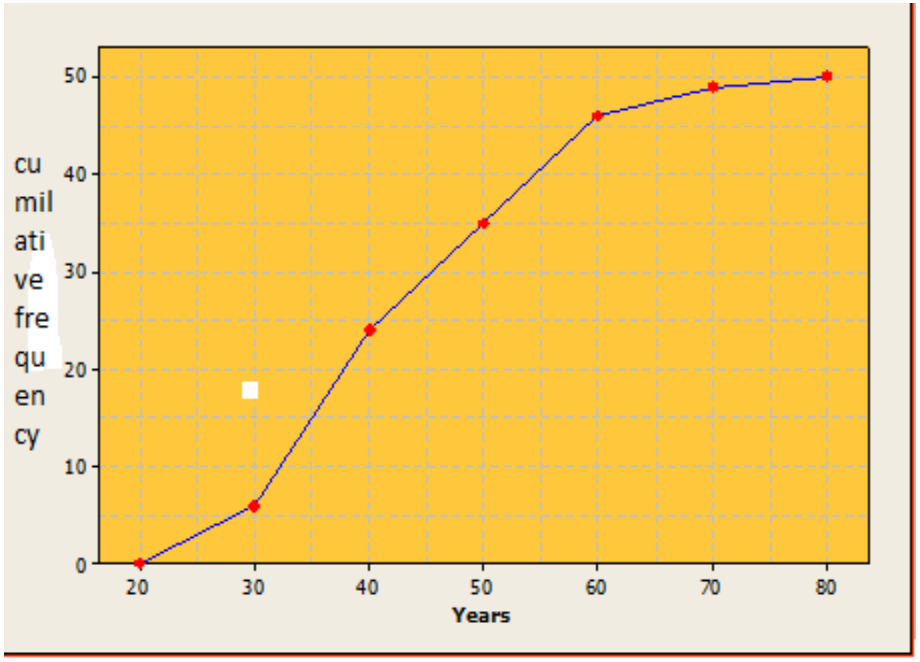
Class Interval	Frequency
20-under 30	6
30-under 40	18
40-under 50	11
50-under 60	11
60-under 70	3
70-under 80	1



Ogive

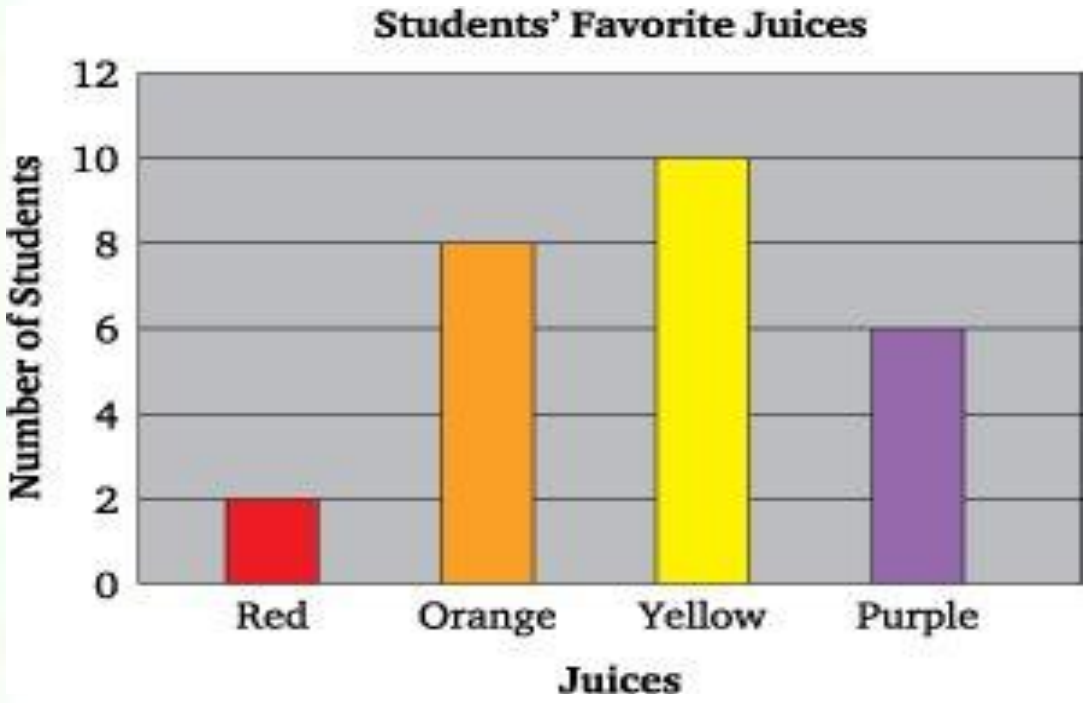
- A graph of a **cumulative frequency distribution.**
- Ogive is used when one wants to determine how many observations lie above or below a certain value in a distribution.
- First cumulative frequency distribution is constructed.
- Cumulative frequencies are plotted at the upper class limit of each category.
- Ogive can also be constructed for a relative frequency distribution.

Class Interval	Cumulative Frequency
20-under 30	6
30-under 40	24
40-under 50	35
50-under 60	46
60-under 70	49
70-under 80	50



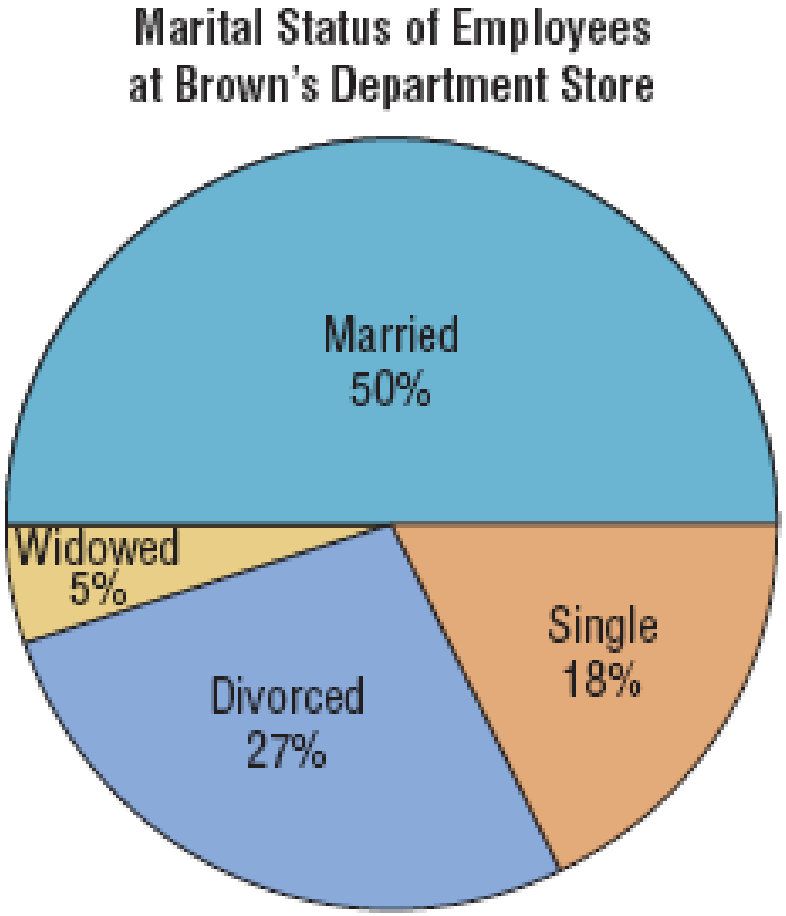
Bar Graphs

- Another common method for graphically presenting **nominal and ordinal scaled data.**
- One bar is used to represent the frequency for each category.
- The bars are separated, and this is why such a graph is frequently used for nominal and ordinal data – the separation emphasize the plotting of frequencies for distinct categories



Pie Graphs

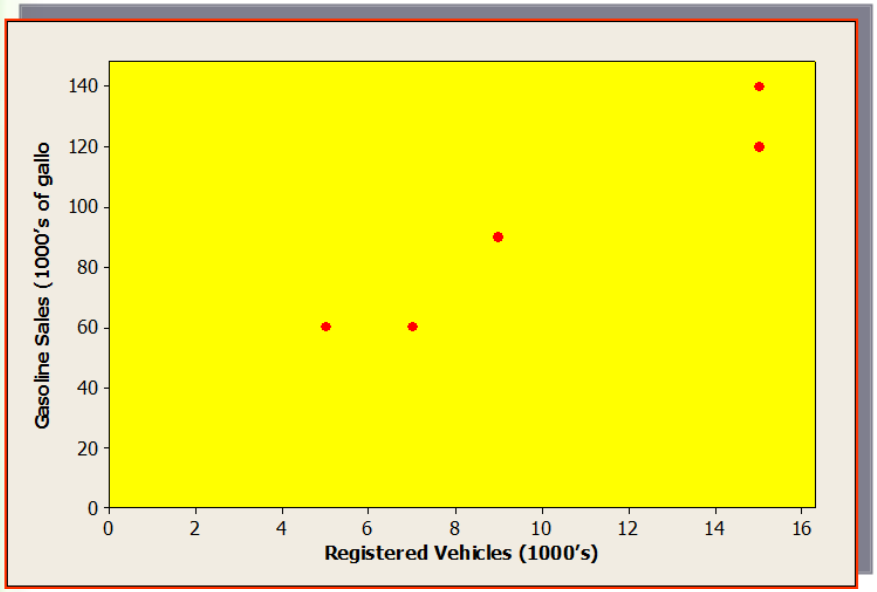
- The pie chart is an effective way of displaying the percentage breakdown of data by category.



Scatter Plot

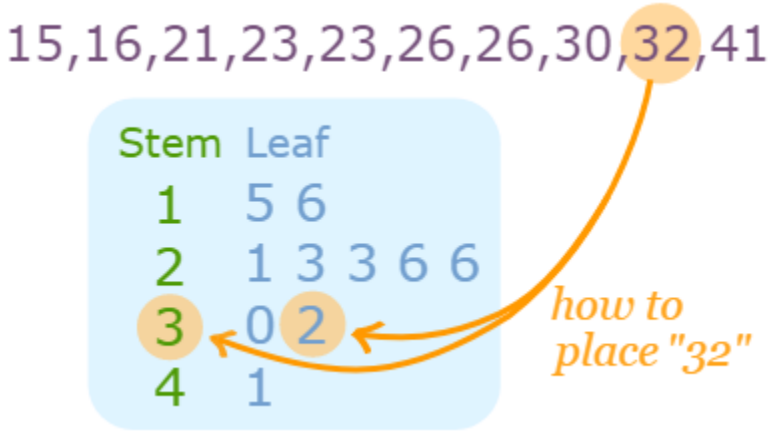
A scatter plot is a two-dimensional graph plot of pairs of points from two numerical variables.

Registered Vehicles (1000's)	Gasoline Sales (1000's of gallons)
5	60
15	120
9	90
15	140
7	60



Stem and Leaf Plots

- A stem and leaf plots is a data plot that uses part of a data value as the stem and part of the data value as the leaf to form groups or classes.
- The "stem" values are listed down, and the "leaf" values go right (or left) from the stem values.
- The "stem" is used to group the scores and each "leaf" shows the individual scores within each group.



- Stem "1" Leaf "5" means **15**
- Stem "1" Leaf "6" means **16**
- Stem "2" Leaf "1" means **21**

At an outpatient testing center, the number of cardiograms performed each day for 20 days is shown. Construct a stem and leaf plot for the data.

25	31	20	32	13
14	43	2	57	23
36	32	33	32	44
32	52	44	51	45

Ordered Stem Plot

0	2
1	3 4
2	0 3 5
3	1 2 2 2 2 3 6
4	3 4 4 5
5	1 2 7

Construct a stem-and-leaf plot for the data in the table.

Cloth Lengths (centimeters)									
5	1	20	12	27	2	30	5	7	38
40	47	1	2	1	32	4	44	33	23

Solution:

Step 1: Sort the data values: 1, 1, 1, 2, 2, 4, 5, 5, 7, 12, 20, 23, 27, 30, 32, 33, 38, 40, 44, 47

Step 2: Choose the stems and the leaves. As the data values range from 1 to 47, use the tens digits for the stems and the ones digits for the leaves. Be sure to include the key.

Step 3: Write the stems to the left of the vertical line from the top to bottom.

Step 4: Write the leaf values corresponding to each stem to the right of the vertical line.

Stem	Plot
0	1 1 1 2 2 4 5 5 7
1	2
2	0 3 7
3	0 2 3 8
4	0 4 7

Key : 0 | 1 = 1 cm



THANK YOU